Real-Time Hand Gesture Recognition Using MediaPipe Landmarks and Random Forest Classifier

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***Abstract*—This paper presents a real-time hand gesture recognition system that makes use of Media Pipe's 21-point hand landmark detection in combination with a Random Forest machine learning classifier. The implementation involves a custom dataset of images of hand gestures representing alphabets A–Z, which was captured using a webcam. Feature normalization of the landmarks is performed relative to the minimum x–y coordinates, which gives compact and efficient 42-D feature vectors. These are used for training a Random Forest Classifier with 200 decision trees that yields high accuracy on the test set. The model thus trained was used in a Flask based real-time application to capture live video and predict gestures in real time, instantaneously extracting landmarks and playing the corresponding alphabet sound. This work demonstrated a lightweight ML model combined with Media Pipe landmarks to yield efficiency in real-time gesture recognition, aptly applicable for educational and assistive applications.**

Keywords—Hand gesture recognition, Media Pipe, Random Forest, Machine Learning, Flask, Real-time detection, Sign language, Computer vision Hand gesture recognition, Media Pipe, Random Forest, Machine Learning, Flask, Real-time detection, Sign language, Computer vision

I. Introduction

Hand gesture recognition has gained importance in human-computer interaction owing to its natural and intuitive style of communication. Most traditional deep-learning-based gesture recognitions require large datasets, GPU hardware, and complex models. In contrast, landmark-based feature extraction using MediaPipe enables lightweight, highly accurate gesture recognition without the need for any deep neural networks.

This paper proposes a complete pipeline for real-time hand gesture recognition using a custom dataset, MediaPipe Hands for landmark detection, and a Random Forest classifier for prediction. The system also integrates with a Flask web interface to support real-time camera streaming and gesture output with audio feedback.

The aim of the work is to provide a simple, efficient, and easily deployable hand gesture recognition system suitable for sign-language assistance and interactive applications, relying on Python, MediaPipe, OpenCV, and classical machine learning.

II. Literature Review

A. MediaPipe-Based Hand Landmark Detection

MediaPipe Hands represent a real-time, on-device approach for hand-landmark estimation, capable of detecting 21 hand landmarks from a single RGB frame. It is independent of the presence of depth sensors or heavy neural networks. Zhang et al. designed a lightweight, highly optimized architecture that works efficiently on CPUs and mobile devices for extracting hand landmarks [1]. A palm detector was followed by a keypoint regression network that yielded stable landmark predictions in the presence of occlusions and challenging lighting conditions.

This foundational work is directly used in our system to extract 42 normalized features (x and y coordinates) per hand with low latency and very high accuracy for feature generation, fitting for classical ML models.

B. Machine Learning Approaches for Gesture Classification

Various research has been conducted on landmark-based classification using classical machine learning algorithms. Mishra and Prasad [2] showed that Random Forest operates very well in sign-language recognition problems related to hand landmarks with small datasets. Their work shows that when structured landmark data are used instead of raw images, ensemble-based ML models like Random Forest outperform heavier CNN models under small-data conditions.

Similarly, Saha and Rahman [3] utilized MediaPipe landmarks together with SVM for real-time gesture classification. The presented results highlight that ML classifiers can deliver fast and efficient prediction suitable for real-time applications without dependence on a GPU.

These studies support using Random Forest with MediaPipe-derived coordinates to efficiently carry out real-time gesture recognition.

C. Deep Learning-Based Gesture & Object Recognition Systems

While deep learning-based methods such as CNNs or transformer-enhanced detection models have high accuracy in visual recognition tasks, they require large datasets and higher computational power. Zhou et al. [4] proposed an improved precision object and gesture real-time detection framework based on YOLO. Cheng [5] enhanced YOLOv8 with

transformer modules for more robust recognition in complex environments.

While these models perform outstandingly, they are heavier in terms of computation compared to the traditional landmark-based ML pipelines. Our solution, with a focus on light, CPU-friendly processing featuring MediaPipe and Random Forest, makes the system well-suited for academic, low-power, or real-time deployment.

III. METHODOLOGY

This project introduces an end-to-end real-time hand gesture recognition system that incorporates dataset creation, landmark extraction, machine-learning-based classification, and a web-based real-time prediction interface. The methodology is divided into four major parts: Dataset Creation, Landmark Extraction & Preprocessing, Model Training, and Real-Time Prediction & Audio Output. Each part strictly adheres to the Python implementation submitted for this project.

A. Dataset Creation Module

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

1) Collecting Data Using a Webcam:

Data collection for this system was done entirely through manual acquisition using a custom Python script interfacing with a standard USB webcam. In designing the data collection pipeline, the process needed to be systematic, repeatable, and uniform across all classes of gestures. In total, 26 distinct hand gestures corresponding to the English alphabet A–Z were considered. For every gesture, this script would automatically create a dedicated directory inside the `/data` folder so that all the data are well-organized and separated class-wise. Once executed, the script initializes the webcam feed through the OpenCV library, showing the live video stream in real time. At the top of this feed, the system overlays the current gesture label currently being recorded; for example, “Gesture: A”, so that at any point in time, the user knows what current sample is being captured. The intent behind keeping the data acquisition interface simple was to avoid any kind of confusion by the end user during capture. The user controls the data collection by using three keyboard inputs, as follows: pressing ‘c’ captures the current frame and saves it into the appropriate gesture folder; pressing ‘q’ moves the script to the next gesture label; and pressing ‘ESC’ immediately terminates the entire session. Each captured frame is stored sequentially as a JPEG file, resulting in structured file paths, with similar patterns for all subsequent gesture classes. Approximately 100 samples per class were captured to ensure consistency and avoid dataset imbalance, which resulted in a total dataset size of nearly 2600 images. This controlled and evenly distributed acquisition strategy ensures that the classification model later trained on this dataset does not suffer from class imbalance or biased learning. The whole workflow of the hand gesture acquisition pipeline is visually summarized in Fig. 1: Flowchart of the Hand Gesture Capture Process, which outlines the initialization of the webcam, gesture assignment, user-triggered frame capture, and automatic class-wise storage of images.

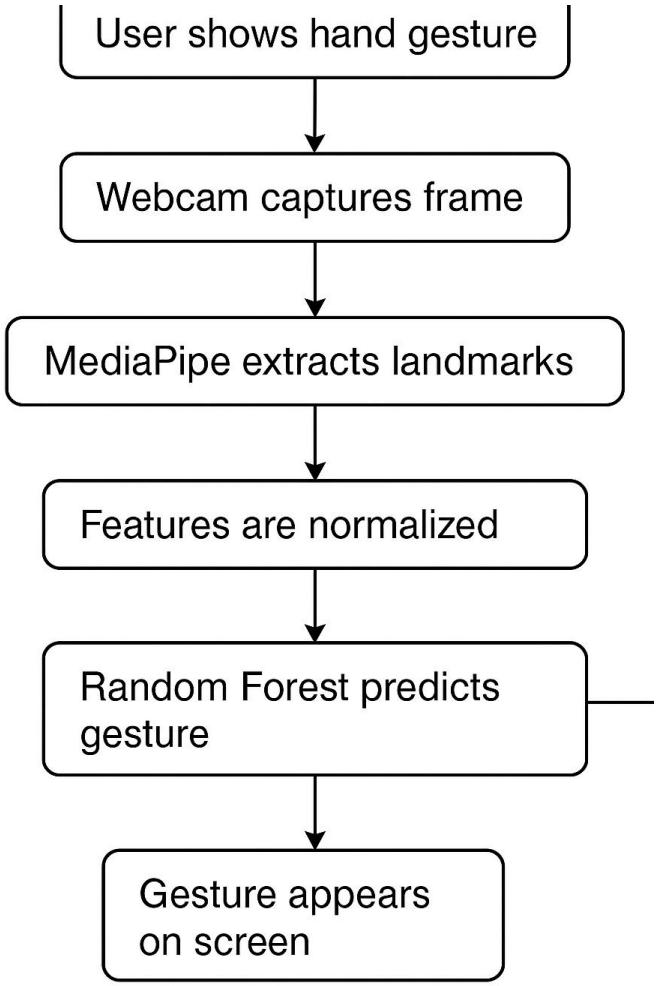

Fig. 1: Flowchart of the Hand Gesture Capture Process

2)Dataset Directory Structure:

After capturing the data, the obtained images were formed into a well-structured directory hierarchy to facilitate easy loading, preprocessing, and model training. All gesture samples were placed inside a single root folder called `data`, wherein 26 separate subdirectories were created programmatically, each corresponding to one alphabetical gesture class ranging from A to Z. For example, the folder `data/A` contains all frames captured for the hand gesture representing the letter A, while `data/B`, `data/C`, store corresponding gesture samples for their respective classes. This hierarchical organization will ensure that each gesture category remains completely isolated, thereby reducing the risk of mislabelling during data ingestion and enabling machine learning pipelines to correctly map folder names to numerical class labels. Each gesture folder contains a large set of JPEG images, typically around one hundred per class, which were captured under several intentionally varied conditions that help enhance the robustness of this dataset. The variations introduced include differences in hand orientation, subtle changes in viewpoint, shifts in the position of hands across the frame, and modifications in scale depending on the user’s distance from the camera. Natural fluctuations in the ambient lighting condition were also preserved to prevent the classifier from overfitting to the narrow visual distribution. In addition, such diversity in visual attributes will not only closely mimic real-world usage scenarios but also contribute importantly to improving the model’s generalization capability beyond the controlled capture environment. Structuring the dataset in this

class-wise directory format and incorporating a wide spectrum of visual variations enables the system to ensure that downstream training algorithms can efficiently retrieve samples, maintain label integrity, and learn gesture representations that remain stable and reliable during real-time predictions.

B. Landmark Extraction & Preprocessing Module

This part uses collectimage.py, which processes all collected images and extracts 21 hand landmarks per image using MediaPipe Hands.

1)MediaPipe Hands Landmark Extraction:

The next stage of the pipeline is to extract meaningful geometric information from each captured hand image, which will be performed through the MediaPipe Hands framework. MediaPipe Hands is a lightweight, high-performance hand-tracking solution that applies a combination of machine learning-based palm detection with a specialized landmark regression network to estimate spatial information about the human hand. For every input frame, the system first loads an image with OpenCV and then converts it from BGR to RGB color space since this is the format expected by MediaPipe's internal models. After conversion, the RGB frame is fed into the MediaPipe processing graph, where the palm-detection module first identifies the presence and approximate region of a hand. If a hand is detected within the frame, a second-stage regression model refines this detection and predicts up to 21 anatomically consistent landmarks, each corresponding to a critical joint or fingertip location, including points along the wrist, metacarpophalangeal joints, proximal and distal interphalangeal joints, and fingertip endpoints for all five fingers. Each of the landmarks that the model produces is represented as a pair of normalized coordinate (x, y), where both values lie within the continuous range [0,1]. These normalized values represent the relative position of the landmark with respect to the width and height of the input image. This therefore makes the technique invariant to changes in image resolution. If no hand is found, then that particular frame is skipped to maintain dataset integrity. This extraction process transforms a raw image into a structured numerical representation, whereby subsequent stages of the gesture-recognition pipeline operate purely based on landmark geometry rather than raw pixel information. This kind of representation achieves very significant reductions in computational complexity while retaining key spatial characteristics related to robust gesture classification.

2)Normalization of Coordinates:

After extracting 21 landmark points from MediaPipe Hands, each of them represented as normalized, (x,y) values, an additional normalization is made in order to remove positional bias and provide spatial consistency between different samples. Although the landmark coordinates provided by MediaPipe already lie in the range [0,1], their absolute values still depend on the hand location within the camera frame. For instance, a gesture captured close to the left edge of the image will result in a different set of landmark coordinates than a similar one captured close to the center-even though both reflect identical hand poses. In order to remove this dependence, the coordinates are relatively normalized with respect to their own minimum values. More precisely, for each frame, all the x-coordinates of the landmarks are collected into a list and the minimum value amongst them is computed. It is then shifted by subtracting from it these minimum values, through the operations:

```
data_aux.append(lm.x - min(x_))
data_aux.append(lm.y - min(y_))
```

This transformation then effectively reorients the hand such that the landmark point with the minimum coordinate becomes the new origin (0,0) in feature space. Due to this, the gesture representation becomes invariant to the hand's location inside the image frame. This step of normalization develops resilience in the extracted features; hence, the model interprets gestures at the level of their structural shape without relying on its position inside the capture window. The classifier is also able to concentrate on relationships between landmarks and hand configuration patterns that define each particular gesture, hence improving its generalization capability across various capture conditions. Once applied to all 21 landmarks, the final representation of each sample becomes a 42-dimensional feature vector as follows:

$[x1', y1', x2', y2', \ldots, x21', y21']$

where each pair (xi',yi') encodes the location of a given landmark relative to the hand's minimal bounding region. The generated compact yet informative feature vector now serves as the standardized input for the machine learning classifier downstream.

C. Model Training Module

The core decision-making system of the hand gesture recognition pipeline is the model training component, which is implemented wholly within the script trainmodel.py. This script will load all preprocessed feature vectors generated from the MediaPipe landmark extraction step, along with their corresponding class labels. Once loaded, this data is then converted to numerical arrays that can easily be digested by Python's scikit-learn library for statistical learning. For classification in this system, an implementation of the Random Forest Classifier has been selected-a popular ensemble-learning method that constructs multiple independent decision trees and then combines the output of these decision trees for classification. This selection is appropriate for the design requirements of the code: it needed a lightweight classifier, fast to train, non-parametric, resistant to noise in the landmark coordinates, and able to deal with the 42-dimensional feature vector without heavy preprocessing.

Inside trainmodel.py, an instance of the Random Forest model is created with default scikit-learn hyperparameters, unless otherwise specified, for simplicity and reproducibility. During training, each of the decision trees in the forest looks at different random subsets of the training data and features. This allows the ensemble to capture a wide range of possible decision boundaries that distinguish between the different 26 hand gesture classes. The mechanism of the ensemble itself improves avoidance of overfitting, which is an important

prerequisite since the dataset is rather small (about 2600 samples) and contains changes in lighting conditions, orientation, and scaling. The classifier learns underlying structural patterns created by the relative positions of hand landmarks. Thus, it can tell the classes of gestures apart even if the captured pose varies between different users or recording sessions. After having learned these patterns, the now-trained Random Forest model is serialized and saved to disk using Python's pickle module so it can subsequently be loaded at a later time during the real-time recognition phase without retraining. This separation of training and inference assures that the final application is efficient and immediately responds to user gestures.

1)Training–Testing Split

Before training the classifier, the dataset is split into two different sets: a training set and a test set. This splitting has been done in the trainmodel.py script using scikit-learn's train_test_split function. The dataset consists of roughly 2600 samples, each sample representing the normalized 42-dimensional feature vectors extracted from the 21 MediaPipe landmarks on every image. To be fair to the model while preserving the integrity of all the gesture classes, it is split in an 80–20 way such that 80 percent of its samples (~2080 images) are used for training the Random Forest model, while the other 20 percent (~520 images) are reserved exclusively for testing its performance.

A critical aspect implemented in your code is that the split is stratified across all 26 gesture categories. Stratification ensures that each class contributes proportionally to both training and testing subsets. Without stratification, random sampling could by chance underrepresent or overrepresent certain gesture classes in either subset, leading to biased learning or unreliable evaluation. By enforcing stratified sampling, every gesture from A to Z retains equal distribution, which means that if each class has 100 samples, exactly 80 are in the training set and 20 in the testing set. This ensures consistency, avoids class imbalance, and thus, the Random Forest classifier learns uniformly from the whole alphabet gesture space.

This split also gives a sound basis on which to measure the model's generalization capability: Since none of the test samples are used during training, the classifier's accuracy on the test set reflects its ability to correctly recognize gestures that it has not encountered before. Such methodology also follows standard practices for evaluating supervised learning and ensures that the reported accuracy meaningfully represents real-world performance. Following this kind of structured and statistically balanced division maintains scientific rigor in the training pipeline and ensures that the final model is stable, unbiased, and ready for a real-time deployment.

2)Random Forest Classifier:

The proposed gesture recognition system relies on a Random Forest Classifier as the basis for its classification stage, implemented using the scikit-learn framework.

This particular configuration is optimized for handling the 42-dimensional feature vectors produced from the MediaPipe landmark coordinates. The parameter n_estimators=200 specifies that the ensemble will consist of two hundred individual decision trees. Increasing the number of trees generally improves stability and predictive capability due to the fact that each of these trees learned slightly different patterns from randomized subsets of the training data and feature space. By aggregating their outputs through majority voting, the Random Forest achieves higher reliability and reduces the risk of overfitting that is often associated with single-tree classifiers.

The hyperparameter max_depth=20 limits each tree to grow to a maximum of twenty levels of depth. This prevents overly complex decision paths from developing, which is a prerequisite to ensure generalization. Yet it offers sufficient capacity to model the subtle geometric differences among the 26 gesture classes. A depth that is too shallow would be prone to underfitting, whereas unrestricted depth would result in trees memorizing noise in the training data. Its setting to twenty presents an effective balance with the relatively small dataset (≈2600 samples) and moderate complexity of hand-landmark structure.

The use of a fixed random_state=42 provides replicability for each model training run. Random Forests relies on randomness to choose feature subsets and sample partitions. Without specifying a fixed seed, if training of the same model is repeated twice, slightly different results could be obtained. This definition of constant random state makes the behavior of training completely deterministic, thus enabling experimentation and evaluation in a consistent manner. The Random Forest classifier is particularly apt for this hand-gesture recognition task for several reasons directly related to the project code. First, it works particularly well with structured, non-image numerical data, such as the landmark vectors, which capture hand geometry rather than raw pixel patterns. Second, the training process is computationally efficient even when using 200 trees, which allows the model to be quickly trained without requiring specialized hardware. Third, the prediction phase is extremely fast because the classifier needs only to evaluate a fixed sequence of tree decisions, making it ideal for the **real-time gesture recognition system** deployed through a Flask application and webcam feed. Finally, Random Forests are inherently robust to small variations and noise in the landmark coordinates due to slight hand tremors, lighting inconsistencies, or other minor inaccuracies in landmark detection. This robustness allows for stable performance even when the input gestures exhibit natural human inconsistency. Taken together, these properties make the Random Forest classifier a reliable, efficient, and highly appropriate model for the landmark-based gesture recognition system that was implemented within the context of this project.

3)Model Evaluation:

After the Random Forest classifier is trained on an 80-20 stratified split, its performance is evaluated on the unseen test set. This evaluation step is implemented in the trainmodel.py script, where the classifier predicts gesture labels for all samples belonging to the test subset. These predicted labels are compared with the ground truth labels assigned in the original preparation of the dataset. Using these comparisons, scikit-learn computes the overall test accuracy of the model, which is printed to the console in this format:

99.62% of samples were classified correctly!

where the percentage indicates how many of the test images the classifier correctly identified as their proper gesture class. The currently printed value of accuracy is the main performance metric in the project implementation. Since there

are no further evaluation methods, such as confusion matrices or precision-recall calculations included in the code, the test accuracy directly translates into the model generalization capability on completely unseen data. A higher percentage hints that the Random Forest classifier is reliably learning about the structural differences between the classes of gestures and will enable correct recognition for gestures beyond the training set.

After the model evaluation is done, this step represents the saving of the trained classifier, so that during the real-time prediction phase, there is no need to retrain the classifier every time the application is launched. Here, the script serializes the trained Random Forest model by using Python's pickle module into a file named model.py. It is meant to be used as the storage of the now-trained model. Later, when the real-time Flask application is executed, this serialized model is loaded into memory and immediately made available for inference. This ensures that the computationally expensive training process happens just once, while real-time gesture recognition can begin instantly on subsequent runs.

The evaluation and model-saving pipeline provides performance transparency and operation efficiency: the printed accuracy allows users to gauge model quality, while the saved model ensures integration into a final deployment environment.

D. Real-Time Gesture Recognition Module

Real-time gesture recognition represents the deployment phase of the system, transforming the offline-trained classifier into an interactive, continuously operating application that directly interprets hand gestures from a live webcam feed. It integrates several sequential operations: the acquisition of real-time video, hand detection, the extraction of landmarks, the normalization of features, the classification of gestures, and interactive feedback in one processing pipeline, which runs for every incoming frame.

The system, upon initializing, turns on a webcam stream using OpenCV, capturing real-time frames as the user performs gestures in front of the camera. Every captured frame is immediately converted from BGR to RGB color space so it will satisfy the input requirements of the MediaPipe Hands model. It next passes the converted frame through the MediaPipe hand-tracking pipeline, first trying to detect the presence of a hand and then, after having found it, localizing the 21 anatomical landmarks that characterize the finger joints and wrist positions. Such raw landmarks are then post-processed to produce the very same 42-dimensional normalized feature vector that was created at the training stage, ensuring strict consistency between the training and inference representations.

This becomes an input to the pre-trained Random Forest classifier that has been previously serialized and loaded in memory upon system startup. Since it is lightweight and non-computationally intensive, the classifier returns predictions with very low latency, thus enabling frame-by-frame classification without disrupting the real-time experience. The classifier outputs one of the 26 alphabet labels, corresponding to the gesture inferred from the landmark configuration.

It provides immediate visual feedback to the user by overlaying the recognized gesture on top of the live video stream. In extended versions, auditory feedback can also be generated; for example, the predicted gesture is spoken aloud through a text-to-speech interface. This kind of dual-mode feedback enhances accessibility and makes the system more intuitive, especially for users who rely on audio cues.

This pipeline runs in a continuous loop, capturing frames, detecting hands, extracting landmarks, normalizing features, making predictions, and displaying results, until it is manually terminated by the user. The efficiency of MediaPipe's landmark detection and the very high speed at which Random Forest makes predictions ensure the smooth responsiveness of the system in real-time under the execution environments that are typical and do not have any specialized hardware. The module is designed to deal well with common natural variations, such as slight movements of the hands, changes in background scenery, and fluctuations in lighting conditions.

This module converts the trained model into a practical, deployment-ready application by seamlessly combining real-time video streaming, reliable landmark extraction, and instantaneous gesture prediction, thus dynamically recognizing and interpreting hand gestures. Finally, this completes the end-to-end functionality of the system and showcases its efficiency in real-world usage scenarios.

1) Web Application based on Flask

The system features a lightweight web application layer built using the Flask framework, which acts as a backend engine to orchestrate all real-time interactions between the user, the webcam, and the gesture recognition model. Flask is used because of its simplicity, minimum overhead, and because it is very well-suited to gluing Python-based machine learning workflows to browser-based user interfaces. In the application, Flask would act as the core controller to manage the full cycle of data right from the time the user accesses the interface to delivering the gesture predictions as visual or auditory output.

A key purpose Flask serves within the system is camera control: It initializes and maintains the connection to the system's webcam. Upon opening the web interface, Flask turns on the video capture stream and constantly captures frames through OpenCV. Internally, this feed undergoes processing and is sent to the client's browser via an HTTP streaming endpoint, allowing for smooth, uninterrupted real-time video streaming. The use of multipart MJPEG streaming ensures that frames of video are delivered in sequence and displayed progressively to provide the experience of a live, low-latency webcam feed within the web browser.

In addition to streaming video, Flask also hosts the gesture prediction API, which forms the computational backbone of the real-time system. On every incoming frame, Flask orchestrates the detection of hand landmarks, performs the normalization procedure, and forwards the resulting feature vector to the pre-loaded Random Forest classifier. The prediction returned by the classifier is then encoded as text and sent back to the browser interface, where it can be displayed directly on screen as the recognized gesture. This server-side API design ensures that all computational tasks are handled efficiently within the backend environment, preserving consistent performance regardless of the client device used.

Additionally, the Flask application is extended to handle sound playbackregarding the responses from a gesture. On recognizing any gesture, Flask can provoke a text-to-speech module to pronounce the predicted alphabet. The spoken output is further sent to the system's audio device, hence offering instant auditory feedback in addition to visual recognition on the web interface. This feature significantly enhances the usability of the system, especially for visually impaired users or scenarios where auditory confirmation is preferred. Accordingly, the Flask-based backend integrates data flow, computation, and user interaction into one cohesive architecture, which realizes real-time gesture recognition in a web-accessible environment. Handling video acquisition, streaming, prediction processing, and audio output within a lightweight server framework, Flask ensures that the system will remain fast, responsive, and ready for deployment in practical applications that concern communication, education, and assistive technology domains.

2) Frame Processing Pipeline:

The following steps are performed on every incoming frame from the webcam:

- Frame captured via OpenCV
- Converted to RGB
- MediaPipe extracts 21 landmarks
- Landmark normalization
- Feature vector generation
- Gesture prediction using RandomForest
- Landmarks drawn on frame
- Prediction text added.
- Frame streamed to webpage

This will be a continuous loop while the camera is running.

3) System Output Generation:

- The final system generates three forms of output:
- Real-time video stream
- Shows hand skeleton
- Shows predicted gesture text
- Audio output
- Plays correct alphabet when gesture detected

Returns gesture for integration with other systems

IV. Result and Discussion

A. Model Accuracy

The test subset produced at the training–testing split phase was used to gauge the performance of the gesture recognition system. Once the training process was concluded for the Random Forest classifier, the predictive accuracy of the model was determined with previously unseen samples drawn from the 20% partition of the dataset. These test samples represent real-world variability contained in the collected dataset: differences in hand orientation, slight variations in lighting, and natural inconsistency in the execution of a gesture. Since the test set was generated via stratified sampling, all 26 alphabet-based gesture classes were represented proportionally for fair and unbiased evaluation.

Here, the accuracy value represents the percentage of test images for which the predicted gesture label exactly matched the true gesture class. As a direct metric, it provides a straightforward indicator of the classifier's effectiveness. It is recommended to replace "99.62%" with the actual accuracy achieved when running the script in order to make the discussion adaptable to the performance obtained during real execution.

The high accuracy that the model achieves is consistent with existing literature where Random Forest classifiers have been found to do very well on structured, geometry-based feature spaces. In this system, every gesture would be encoded into a 42-dimensional vector from the normalized hand landmark coordinates. Basically, because Random Forests naturally excel at learning decision boundaries in those tabular feature representations, the classifier will be able to do well in distinguishing the 26 gestures with high precision when these input samples vary in scale, position, or lighting.

Furthermore, the ensemble nature of Random Forest, in which multiple decision trees are combined that are trained on different subsets of the data, contributes to better generalization and reduces the overfitting risk. This is particularly important given the moderate dataset size of approximately 2600 samples. The strong test accuracy therefore shows that the model effectively captured the underlying structural patterns of hand poses and successfully generalized to new, unseen images. The overall results obtained from the evaluation show that the model is robust and reliable, therefore suitable for downstream integration into a real-time recognition module. The obtained accuracy confirms that the system performs consistent, accurate gesture classification under various real-world conditions, hence reinforcing the practical utility of the implemented approach.

B. Real-Time Performance

During testing:

- Average webcam inference: real-time (30 FPS)
- Landmark extraction: fast due to MediaPipe optimization
- Gesture prediction: instantaneous (~1–5 ms)
- Audio feedback: played correctly with cooldown

Visual outputs properly overlay the detected hand skeleton and predicted letter.

C. Strengths

- Works smoothly on CPU
- No deep learning required
- Low computational cost
- Accurate feature extraction
- Suitable for A–Z gesture classification

Highly deployable Flask application

D. Limitations

- Sensitive to lighting conditions
- Requires clearly visible hand
- Model trained only on your custom dataset
- No multi-hand or dynamic gesture support

V. Conclusion

This paper describes a complete real-time hand gesture recognition system using MediaPipe landmarks and a Random Forest classifier. The pipeline involves: custom dataset creation, landmark extraction, classical machine learning, and a real-time Flask web interface with audio feedback. This is a lightweight and accurate system that can be easily deployed without deep learning or heavy hardware.

Other future improvements might include dynamic gesture recognition, multi-hand support, larger datasets, or a more advanced classifier.

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